

Pre-work · Urinary and Endocrine

The Urinary System and Endocrine Anatomy. Read both Part A chapters first. The two chapters share a sheet because an adrenal gland caps each kidney and the kidney is itself an organ with endocrine tissue, so learning them together saves you a second pass over the same region.

Module 4 · 3 of 5

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

- 1 Read**
Read the Part A chapter first.
This packet does not repeat it. It puts it to work.
- 2 Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
- 3 Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
- 4 Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
- 5 Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the urinary and endocrine chapters into mini visuals. Eight chunks. The kidney takes six of them, because it is the one organ in this module you have to hold at three scales at once.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 LADDER

The kidney from the outside in

Five rungs, renal fascia at the top down to the medulla. Beside each covering write the one job it does. On the last two rungs draw the pyramids with their bases toward the cortex and the renal columns dipping down between them.

2 HUB

The kidney in frontal section

One bean at the hub, hilum on the medial side. Spoke to the cortex, the columns, the medulla, the pyramids, the papillae, the minor and major calyces, the renal pelvis, and the ureter. Draw each papilla pointing into its own minor calyx, because that is the joint most people draw backward.

3 CHAIN

The blood supply, aorta to renal vein

Eight boxes with arrows, renal artery through to the renal vein. Circle the two capillary beds and box the single vessel that sits between them, then write beside each bed what happens there.

4 HUB

The renal corpuscle and the juxtaglomerular apparatus

One corpuscle at large scale in the middle. Spoke to the glomerulus, the glomerular capsule with its two layers, the podocytes and their filtration slits, and the capsular space. Then draw the afferent and efferent arterioles and the distal tubule curving back to touch them, and label the macula densa and the juxtaglomerular cells at that contact.

DRAWING 1

The kidney in frontal section, with its vessels

BRAIN DUMP FIRST

Two minutes. Every region, every collecting space, and the arterial chain running through them.

Draw a kidney cut front to back, hilum on the medial side. Lay in the cortex, the medulla with several pyramids, the renal columns between them, and the papilla at the tip of each pyramid. Draw the minor calyces cupping the papillae, the major calyces, the renal pelvis, and the ureter leaving at the hilum. Then draw the arterial chain over the top of it: renal artery at the hilum, segmental and interlobar arteries running in the columns, arcuate arteries arching along the bases of the pyramids, and interlobular arteries branching up into the cortex.



In your notebook, head the page **M4-1 The kidney in frontal section, with its vessels** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can put a finger anywhere in your kidney and say whether the fluid there is still inside a tubule or already in a collecting space.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A urine sample contains albumin, a large plasma protein, but no blood cells. Name the layer of the filtration membrane most likely damaged and explain what rules out the other two.

Answer. The basement membrane.

Reasoning. Read the sample as a sieve test rather than as a disease. Three layers stand between the blood and the capsular space, and each one holds back something of a different size, so what escaped names the layer that failed. The fenestrated endothelium of the glomerular capillary is the layer that stops blood cells, and no cells appear in the sample, so that layer is still doing its job. The slit membranes between the podocyte processes hold back medium-sized proteins, and had they alone failed you would expect smaller proteins through first, not the largest one. Albumin is a large plasma protein, and the layer built to hold back large plasma proteins is the basement membrane lying between the capillary and the podocytes. The size of what got through is the whole of the reasoning.

Set A · Name it

1. The cup-shaped chamber that collects urine draining from a renal papilla is a _____.
2. The cells of the afferent arteriole that release renin when blood pressure is low are the _____.
3. The smooth triangular area at the base of the bladder, marked by three openings, is the _____.
4. The bridge of thyroid tissue crossing the front of the trachea is the _____.
5. Name the three zones of the adrenal cortex in order, from the capsule inward.

5 LADDER

The three layers of the filtration membrane

Three rungs, in the order plasma crosses them. Beside each rung write what it lets through and what it holds back. The rungs get finer as you go down, and that gradient is the whole answer.

6 CHAIN

From capsular space to the outside

One long chain with arrows: capsular space, the three tubule segments, the collecting duct, then papilla, calyces, pelvis, ureter, bladder, urethra. Draw a line across the chain at the point where the fluid stops being called filtrate and starts being called urine.

7 GRID

Where every endocrine gland sits

Gland down the side, ten rows, running from the top of the body to the bottom. Location and one note across the top. Add a third column: is this a pure endocrine gland, or an organ with a second job.

8 SPLIT

The two lobes of the pituitary

One split. Other name, tissue, and link to the hypothalamus on each branch. The tissue column predicts the link column, so draw an arrow between those two cells on both branches.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

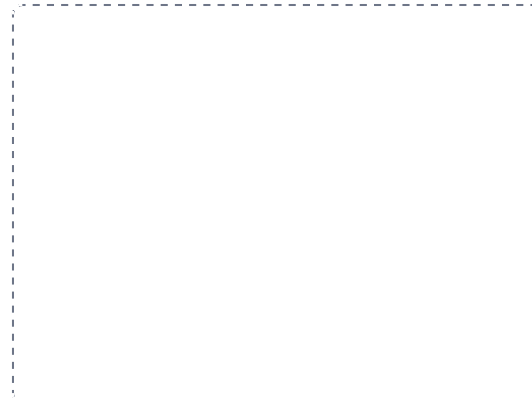
DRAWING 2

One nephron, drawn in place

BRAIN DUMP FIRST

Two minutes. Every segment in order, and which region of the kidney each one sits in.

Draw a wedge of kidney with the cortex above and the medulla below, and the line between them marked. Put a renal corpuscle in the cortex and run the tubule out of it: proximal convoluted tubule coiled in the cortex, nephron loop diving into the medulla and turning back, distal convoluted tubule coiled in the cortex again, then a collecting duct running down the pyramid to the papilla. Wrap peritubular capillaries around the cortical tubules and the vasa recta around the loop. Then draw a second nephron beside the first with a long loop reaching deep, and label which type each one is.



In your notebook, head the page **M4-2 One nephron, drawn in place** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say for every segment of your nephron whether it lies in the cortex or the medulla, and which of the two nephron types concentrates the urine.

Set B · Predict it

1. A patient loses the fat surrounding one kidney. Predict what happens to the position of the organ and to the flow of urine.

2. A pituitary tumor enlarges. Predict the direction it grows and the first symptom, working from the bone around it.

3. A stone forms in the renal pelvis and stays there. Predict when the patient first feels pain and explain why.

4. The stalk between the hypothalamus and the pituitary is cut. Predict which lobe is affected more directly, working from the link each lobe uses.

Set C · Tell it apart

1. Distinguish a cortical from a juxtamedullary nephron by loop length, blood supply, and job.

2. The kidney appears on both the urinary sheet and the endocrine sheet. Defend both placements.

3. Explain why urinary tract infections are more common in women, using one structural difference.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

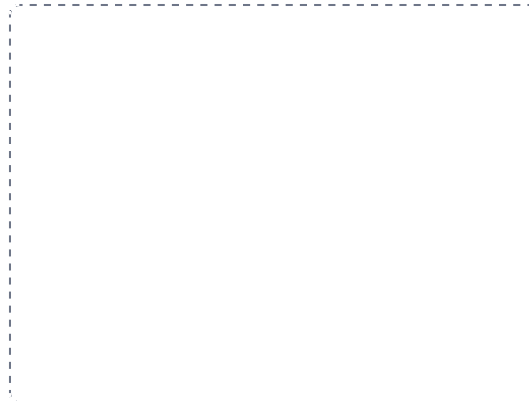
DRAWING 3

The endocrine glands on a body outline

BRAIN DUMP FIRST

Two minutes. Every gland, where it sits, and what the two-part glands are built from.

Draw a body outline from head to pelvis. Mark the hypothalamus and the pituitary hanging beneath it in the sella turcica, the pineal gland at the back of the third ventricle, the thyroid across the trachea with its isthmus and the four parathyroids behind it, the thymus behind the sternum, an adrenal gland capping each kidney, the pancreas behind the stomach, and the ovaries or the testes. Then enlarge two glands in the margin: the pituitary with its anterior and posterior lobes and the infundibulum joining it to the hypothalamus, and the adrenal with its three cortical zones in order and the medulla inside them.



In your notebook, head the page **M4-3 The endocrine glands on a body outline** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name every gland on the outline with the labels covered, and say for the pituitary and the adrenal which part is nervous tissue and which is glandular.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The kidney from the outside in

LADDER

2 The kidney in frontal section

HUB

3 The blood supply, aorta to renal vein

CHAIN

4 The renal corpuscle and the juxtaglomerular apparatus

HUB

5 The three layers of the filtration membrane

LADDER

6 From capsular space to the outside

CHAIN

7 Where every endocrine gland sits

GRID

8 The two lobes of the pituitary

SPLIT